



Medical Policy

Subject: Percutaneous Spinal Surgery

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Description/Scope

This document addresses percutaneous spinal discectomy and disc decompression procedures, image-guided lumbar decompression procedures, and percutaneous posterior cervical decompression/fusion procedures.

During a percutaneous image-guided spinal procedure, the surgeon does not have direct visualization of the anatomic site with the naked eye. Visual guidance is provided indirectly using fluoroscopy.

During open and minimally invasive procedures, the surgeon is able to directly visualize the operative site. This document does not address procedures performed under direct visualization, including open, microscopic/tubular, or endoscopic techniques.

Note: Please see the following related documents for additional information:

- [SURG.00052 Percutaneous Vertebral Disc and Vertebral Endplate Procedures](#)
- [SURG.00073 Epiduroscopy](#)
- [SURG.00111 Axial Lumbar Interbody Fusion](#)
- [SURG.00134 Interspinous Process Fixation Devices](#)

Note: For a high-level overview of this document, please see [Summary for Members and Families](#) below.

Position Statement

Investigational and Not Medically Necessary:

Percutaneous spinal surgical techniques are considered **investigational and not medically necessary**.

Summary for Members and Families

This document describes clinical studies and expert recommendations as to whether certain spinal surgery procedures are clinically appropriate. The following summary does not replace the medical necessity criteria or other information in this document. The summary may not contain all of the relevant criteria or information. This summary is not medical advice. Please check with your healthcare provider for any advice about your health.

Key Information

Percutaneous spinal surgery includes procedures that treat back or neck problems by placing tools through the skin, using imaging like X-ray for guidance instead of directly viewing the area. These procedures aim to remove or shrink parts of a spinal disc or relieve pressure on nerves. Common types include automated percutaneous lumbar discectomy, percutaneous laser disc decompression, and nucleoplasty. Another method, called image-guided lumbar decompression, removes small amounts of tissue to ease spinal canal narrowing. These procedures are less invasive than open surgery.

What the Studies Show

These procedures try to reduce pressure in the spine by removing or shrinking disc material or nearby tissue. They use imaging instead of direct sight, which may limit precision. Some methods use suction, others use heat, laser, or radiofrequency energy (a type of heat-based treatment) to break down tissue. Image-guided lumbar decompression removes small pieces of bone or ligament to create more space in the spinal canal.

Studies comparing these procedures to standard surgery or non-surgical care show mixed or weak results. In several trials, people who had percutaneous procedures needed follow-up surgeries more often than those who had standard surgery. Pain and function improved in some people, but results were often similar to other treatments. Many studies had small sample sizes, no comparison group, or short follow-up. Systematic reviews determined that better studies are needed to know if these procedures improve health. Some studies also raised concerns about risks such as repeat surgery, disc damage, or device problems.

Is this clinically appropriate?

These procedures are not clinically appropriate because they have not been proven to improve health.

The available research studies have important limits. Many did not compare these procedures to standard care, or they included small groups of people. Some studies showed higher rates of repeat surgery or unclear long-term benefit. Reviews of multiple studies found that current evidence is not strong enough to show these procedures work as well as existing treatments.

Using treatments that are not proven exposes people to risks without a clear expectation of benefits. These risks include a possible need for repeat procedures, worsening disc damage, or complications from devices. More high-quality studies are needed to understand if these procedures are safe and effective.

[\(Return to Description/Scope\)](#)

Rationale

Summary

The available evidence for percutaneous spinal discectomy and disc decompression consists predominantly of uncontrolled case series and small comparative studies with significant methodological limitations, including lack of randomization, selection bias, high attrition, and reliance on subjective outcome measures without blinded assessment. Randomized and comparative studies across multiple techniques, including automated percutaneous lumbar discectomy (APLD), percutaneous laser disc decompression (PLDD), nucleoplasty, and intradiscal radiofrequency-based devices, have not demonstrated outcomes at least as beneficial as conventional microdiscectomy or conservative management and in some cases suggest higher rates of reoperation, raising concerns regarding durability of effect. Trials comparing percutaneous techniques to conservative management are limited by non-equivalent study designs and do not permit valid comparative effectiveness conclusions. Cochrane systematic reviews determined that the evidence for the percutaneous techniques reviewed is insufficient. For image-guided minimally invasive lumbar decompression (MILD) for lumbar spinal stenosis (LSS), randomized evidence demonstrates greater short- to intermediate-term improvement compared with epidural steroid injection. However, interpretation of these findings is limited by lack of blinding, potential sources of performance and expectation bias, attrition, and use of a comparator that does not represent definitive surgical management.

Although clinical practice guidelines from the American Society of Pain and Neuroscience (ASPN) have assigned favorable recommendation grades to MILD and to percutaneous image-guided lumbar decompression (PILD), the supporting trials primarily compare these procedures to epidural steroid injection rather than to established surgical alternatives, limiting conclusions regarding their role in the treatment pathway. For percutaneous posterior cervical fusion using an expandable cage system, evidence is limited to small uncontrolled case series and does not establish comparative effectiveness or long-term outcomes.

Discussion

Percutaneous Disc Decompression Compared to Conventional Surgery or Conservative Management

Revel (1993) reported on a randomized study comparing chemonucleolysis and APLD in which 61% of individuals treated with chemonucleolysis reported favorable results compared to 44% of those treated with APLD. Chatterjee (1995) reported a randomized study comparing APLD with open surgical microdiscectomy in which 29% of individuals in the APLD group reported satisfactory results compared to 80% in the microdiscectomy group.

In a multicenter randomized controlled trial, Brouwer (2015) compared PLDD to traditional microdiscectomy for sciatica from lumbosacral disc herniation. Among 115 participants randomized 1:1, the Roland Morris Disability Questionnaire showed non-inferiority of the laser group at 8 weeks and 52 weeks. However, 24 participants (44%) in the laser group underwent additional surgery during the first year compared to 9 (16%) in the surgery group. At 2-year follow-up (Brouwer, 2017), no significant difference in functional outcomes was found between groups, but the reoperation rate for the laser group was 52% compared to 21% in the surgery group. The authors concluded that PLDD followed by surgery when symptoms were not relieved had comparable outcomes to microdiscectomy alone, and that further comparative studies were needed.

Seddighi (2025) conducted a small, single-center randomized controlled trial comparing PLDD with open surgery for radicular sciatic pain with 2-year follow-up. At 2 years, both groups demonstrated improvements in pain and functional outcomes, with no statistically significant differences between groups; however, the study was not powered to detect clinically meaningful differences. Reoperation occurred in 31% of individuals in the PLDD group compared to 19% in the open surgery group. Although this difference in reoperation rates was not statistically significant ($p=0.314$), the numerical difference is clinically relevant. The study population was limited to individuals with contained disc herniations, which may limit generalizability. These findings do not establish equivalence between PLDD and open surgery and suggest potential concerns regarding durability of PLDD.

Policicchio (2025) conducted a retrospective, single-center observational study evaluating PLDD in individuals with contained lumbar disc herniation who had persistent symptoms following an initial course of conservative therapy. All

participants received conservative treatment prior to PLDD, and outcomes were assessed sequentially rather than in independent comparison groups. While pain scores improved following PLDD (approximately 30% reduction in Visual Analog Scale (VAS) at 6 months), no significant improvement was observed after conservative therapy alone. However, the study design precludes valid comparative conclusions, as the PLDD group was limited to individuals who failed conservative management, and outcomes may reflect cumulative treatment effects, natural history, and regression to the mean. Additional limitations include retrospective outcome assessment with potential recall bias and short-term follow-up. These findings should be considered hypothesis-generating and insufficient to establish comparative effectiveness.

Percutaneous Disc Decompression: Head-to-Head Comparisons of Investigational Techniques

Several studies have compared percutaneous techniques to each other without including a standard-of-care surgical comparator. Guo (2025) conducted a small, retrospective cohort study comparing transforaminal endoscopic lumbar discectomy (a direct-visualization endoscopic technique outside this document's scope) to coblation nucleoplasty combined with collagenase chemonucleolysis (a percutaneous technique within this document's scope) in individuals with lumbar disc herniation and grade I degenerative spondylolisthesis. Both groups demonstrated statistically significant improvements from baseline. While early postoperative outcomes favored the endoscopic group, no statistically significant differences were observed at later follow-up time points. Interpretation is limited by the nonrandomized design, small sample size, and absence of a standard-of-care comparator.

Hu (2025) conducted a retrospective, nonrandomized cohort study of 67 individuals comparing posterior percutaneous endoscopic cervical discectomy (PPECD), a direct-visualization technique, to percutaneous cervical nucleoplasty (PCN) for single-level soft cervical disc herniation. Treatment allocation was based on individual preference, introducing potential selection bias. Both groups demonstrated statistically significant improvements in Neck Disability Index and pain scores from baseline; however, PPECD was associated with modestly lower arm pain scores at follow-up, while no significant differences were observed in disability or overall clinical success rates. The study is limited by its retrospective design, small sample size, short follow-up, and lack of a standard-of-care comparator, and does not establish comparative effectiveness of the percutaneous technique.

Wang (2025) conducted a retrospective, propensity score-matched cohort study comparing PCN combined with ultrasound-guided pulsed radiofrequency (PRF) of the cervical nerve root to PRF alone for cervical radicular pain. Both groups demonstrated significant improvements from baseline. The combination group showed statistically greater improvements in pain and functional scores at later follow-up. However, there was no significant difference in global outcomes based on modified MacNab criteria. The PCN plus PRF group exhibited greater reduction in disc height at 12 months, a finding consistent with the ablative mechanism of the procedure and not associated with reported adverse clinical outcomes. Interpretation is limited by the nonrandomized design, small matched sample, and lack of a standard-of-care comparator.

Li (2025b) conducted a retrospective, nonrandomized cohort study comparing low-energy semiconductor percutaneous laser disc decompression (n=30) to percutaneous endoscopic lumbar discectomy (PELD) (n=60) in adolescents with lumbar disc herniation. Both groups demonstrated improvement in pain and function. However, PELD showed statistically superior outcomes on VAS and Oswestry Disability Index at all follow-up intervals. Interpretation is limited by significant methodological weaknesses, including small sample size, lack of randomization, non-contemporaneous comparator groups, and short follow-up.

Percutaneous Disc Decompression: Uncontrolled Evidence

Early evidence on percutaneous disc decompression and nucleoplasty is derived primarily from small, uncontrolled or nonrandomized studies with consistent within-group improvements but limited ability to establish comparative effectiveness. An uncontrolled case series of 50 individuals treated with the DeKompressor device (Stryker Instruments, Kalamazoo, MI) reported > 70% pain reduction in most participants, but lacked randomization or a comparator (Amoretti, 2006). A prospective nonrandomized cohort of 67 individuals undergoing nucleoplasty demonstrated statistically significant improvements in quality of life at 3-6 months, representing a methodological strength in prospective data collection but still lacking a control group and with limited follow-up (Gerszten, 2006). Similarly, several case series involving cervical (n=50) and lumbar populations (n=69; n=29) reported high rates of pain relief or reductions in analgesic use and disability, with some studies demonstrating statistically significant VAS improvements (Al-Zain, 2008; Calisaneller, 2007; Nardi, 2005). However, these studies were limited by small sample sizes, absence of control groups, and short-term follow-up. Notably, several of these study authors acknowledged that the clinical value of nucleoplasty remained unproven and emphasized the need for randomized, placebo-controlled trials. Collectively, these studies suggest potential symptomatic improvement but are uniformly limited by high risk of bias, lack of comparators, and insufficient long-term outcomes, precluding conclusions regarding efficacy relative to standard care.

Fan (2025) conducted a small, single-center, uncontrolled observational study of 30 individuals with cervical spondylotic radiculopathy who underwent PCN, evaluating the relationship between diffusion tensor imaging (DTI) fractional anisotropy and clinical outcomes. A strong negative correlation was found between fractional anisotropy and Neck Disability Index both pre- and post-procedure; however, no significant correlation was observed with pain (VAS) scores. The study is limited by its small sample size, single-arm design, and reliance on correlational analyses of a surrogate imaging endpoint without a comparator group. These findings are hypothesis-generating and do not establish the clinical utility or prognostic validity of DTI metrics following PCN.

Zhang (2025) reported 5-year results for a single-center retrospective cohort study of 118 individuals treated with coblation nucleoplasty for cervical (n=55) or lumbar (n=63) disc herniation. Participants experienced within-group improvements in pain scores and generally favorable MacNab outcomes. However, reoperation occurred in a subset of individuals (approximately 11% cervical and 17% lumbar). The study is limited by its retrospective design, lack of a comparator group, and potential selection and attrition biases, including exclusion of individuals requiring reoperation from some analyses. As an uncontrolled study, the findings do not establish the effectiveness of coblation nucleoplasty relative to standard care or alternative interventions.

Li (2025a) conducted a single-center retrospective cohort study of 66 individuals with cervical chest pain undergoing PCN, comparing outcomes between those who received a preoperative ultrasound-guided selective cervical nerve root block (SCNRB) to guide target selection and those treated based on imaging alone. Both groups improved from baseline. However, the study reported that individuals with positive nerve block tests had significantly better PCN outcomes than those with negative tests, suggesting a potential selection role for the nerve block test. This study does not establish the independent predictive validity of the nerve root block for selecting candidates likely to benefit from PCN. Interpretation is limited by the nonrandomized design, small sample size, and confounding, as the SCNRB test directly influenced procedural targeting.

Takaoka (2026) conducted a small retrospective comparative cohort study (n=42) evaluating two percutaneous intradiscal decompression devices (Disc-FX®, Elliquence, LLC, Baldwin, NY) compared with L'DISQ (U&I Corporation, Uijeongbu-si, Korea) in individuals with lumbar disc herniation. Both devices showed within-group improvement in pain (numerical rating scale) at 12 months, with no statistically significant difference between them. A secondary analysis showed greater improvement in quality of life measured by the EuroQol 5-Dimension 5-Level (EQ-5D-5L) instrument with Disc-FX. However, this finding was exploratory and not robust in adjusted analyses. The study is limited by its retrospective design, small sample size, lack of randomization, baseline imbalances suggesting selection bias, unblinded outcome assessment, and absence of a standard-of-care comparator, precluding conclusions regarding comparative effectiveness or clinical benefit.

Several studies have reported cautionary findings or limited evidentiary value. Gazzeri (2025) conducted a retrospective radiological cohort study of disc size changes measured using magnetic resonance imaging (MRI) after PLDD. Disc height and volume changes did not demonstrate consistent decompression on MRI, and a subset showed late VAS score deterioration. The study reported statistically significant reductions in measured disc area; however, the study lacked a control group, included only short-term imaging follow-up (2 months), and relied on surrogate radiographic outcomes without establishing correlation to durable clinical benefit. Rybaczek (2025) reported 84-month follow-up data for the Disc-FX procedure in a single-center cohort of 49 individuals with contained disc herniation. Attrition was 89.8% from baseline enrollment to the 84-month assessment, suggesting selection bias among those remaining under observation and severely limiting interpretability. Among those with available data, 26.4% underwent reoperation. Ranc (2025) prospectively studied participant experience following computed tomography (CT)-guided percutaneous lumbar discectomy under local anesthesia, focusing on tolerability rather than comparative efficacy. The study confirmed procedural tolerability but did not provide controlled outcome data.

Systematic Reviews and Meta-Analyses

In a 2007 Cochrane review of 40 randomized controlled trials of surgical interventions for spinal disc disease, Gibson and Waddell (2007) found that microdiscectomy gives broadly comparable results to standard open discectomy. The review concluded that there was a lack of evidence to establish the efficacy and safety of automated percutaneous discectomy, coblation therapy, and laser discectomy.

In a 2014 Cochrane review, Rasouli compared minimally invasive discectomy (MID) procedures to microdiscectomy or open discectomy, including 11 randomized or quasi-controlled trials. MID procedures were found to be potentially inferior for relief of leg pain and lower back pain. High risk of bias was identified in 7 of 11 included studies. The authors recommended additional research to define appropriate indications for MID.

A 2025 Cochrane systematic review of nucleoplasty for cervical radicular pain identified only four randomized controlled trials (n=259), all of which were judged to have a high risk of bias and provided low to very low certainty evidence. Compared with conservative management, nucleoplasty may provide greater short-term pain reduction, but does not appear to improve function or quality of life. When compared with other active interventions, including pulsed radiofrequency and surgical discectomy, the available evidence shows little to no difference in clinical outcomes, with substantial uncertainty due to small sample sizes, imprecision, and methodological limitations. No serious adverse events were reported; however, the overall evidence base is limited, lacks robust sham-controlled trials, and does not establish durable benefit or comparative effectiveness compared to standard surgical care. Overall, the evidence is insufficient to support the routine use of nucleoplasty for disc-related radicular pain (de Rooij, 2025).

Image-Guided Minimally Invasive Lumbar Decompression (MILD) for Spinal Stenosis

The early evidence base for image-guided minimally invasive lumbar decompression (MILD) consisted primarily of small uncontrolled studies with short-term follow-up, alongside a limited number of randomized comparisons to epidural steroid injection (ESI). Retrospective and prospective cohort studies (Chopko, 2010; Lingreen, 2010) reported improvements in pain and function with low rates of major adverse events, but were limited by lack of control groups and short follow-up. A

small double-blind randomized trial (Brown, 2012) found greater short-term improvement in VAS and Oswestry Disability Index (ODI) with MILD versus ESI, although follow-up was limited and crossover of control participants reduced interpretability. Longer-term observational data (Chopko, 2013) suggest sustained improvements in pain and function at 2 years without major safety signals, but remain uncontrolled and subject to attrition bias.

The MiDAS ENCORE trial (Benyamin, 2016) is the largest randomized evaluation of MILD and provides moderate-quality comparative evidence against an active interventional control. It was a prospective, multicenter randomized controlled trial conducted across 26 U.S. sites, enrolling 302 Medicare beneficiaries with symptomatic lumbar spinal stenosis and neurogenic claudication refractory to conservative therapy. Participants were randomized 1:1 to MILD or ESI, with predefined inclusion criteria requiring imaging-confirmed ligamentum flavum hypertrophy and functional limitation. The primary endpoint was the proportion of ODI responders (≥ 10 -point improvement), with secondary endpoints including numeric pain rating scale (NPRS) and Zurich Claudication Questionnaire (ZCQ) domains, all assessed using validated thresholds for clinical importance. At 1 year, the study demonstrated statistically significant superiority of MILD over ESI across all measured outcomes. ODI responder rates were 58.0% for MILD compared to 27.1% for ESI ($p < 0.001$), with similar advantages observed for pain and functional domains. Mean improvements were also greater in the MILD group (for example, ODI improvement ~ 16 points vs. ~ 4 -6 points for ESI), and responder analyses across ZCQ domains consistently favored MILD. Safety outcomes were comparable, with low rates of device- or procedure-related adverse events in both groups. However, several design and conduct features limit interpretation. Blinding was not feasible due to procedural differences, introducing risk of performance and expectation bias. The comparator (ESI) is itself a variable, non-curative intervention with heterogeneous administration (up to four injections allowed), limiting its role as a robust standard-of-care benchmark. Attrition and treatment crossover further complicate interpretation. A meaningful proportion of individuals withdrew or received non-protocol therapies, and nonresponders were imputed in analyses, which may bias effect estimates. Additionally, adjunctive therapy restrictions and inclusion of individuals with prior ESI exposure may reduce generalizability and potentially disadvantage the control arm. The study was also industry-sponsored, with investigator involvement in trial oversight, raising potential concerns for sponsorship bias. Strengths include the randomized multicenter design, relatively large sample size for this intervention, use of validated and clinically meaningful outcome thresholds, and inclusion of an active comparator rather than sham or no treatment. Nonetheless, the trial does not address comparative effectiveness versus surgical decompression or other definitive treatments, and the report by Benyamin was limited to 1-year follow-up data.

The 2-year follow-up of MiDAS ENCORE (Staats, 2018) extends the original randomized trial but introduces important interpretive limitations. The study began as a multicenter randomized controlled trial comparing MILD ($n=143$ treated) to ESI ($n=131$ treated), with outcomes assessed at 6 months and 1 year. However, the randomized comparison was not maintained beyond 1 year, and only the MILD arm was followed to 2 years. As a result, the 2-year findings represent an uncontrolled longitudinal extension rather than a comparative effectiveness analysis. At 2 years, among 99 individuals available for follow-up (from an initial 143 treated with MILD), statistically significant and clinically meaningful improvements from baseline were reported across all measures, including ODI (mean improvement 22.7 points), NPRS (3.6-point reduction), and ZCQ domains (improvements ~ 0.8 -1.0). Responder rates were approximately 72% for ODI and NPRS, and participant satisfaction exceeded clinically meaningful thresholds. These results suggest durability of effect over time, with outcome trajectories appearing stable between 6 months, 1 year, and 2 years. Reported reintervention rates included 5.6% undergoing subsequent surgery and 15.4% receiving additional injections or nerve blocks. However, the evidentiary strength of these long-term findings is limited by several factors. Most importantly, the absence of a control group beyond 1 year precludes assessment of whether observed improvements reflect treatment effect, natural history, regression to the mean, or ongoing adjunctive care. Attrition is substantial: of 143 treated individuals, only 99 (approximately 69%) were included in the 2-year analysis, with exclusions due to withdrawal, additional interventions, loss to follow-up, or death, introducing potential attrition bias and selective outcome reporting. The use of a modified intent-to-treat approach including only observed data at each time point further increases risk of bias if missingness is not random. Additionally, individuals who required alternative interventions were removed from analysis, which may preferentially exclude poorer outcomes. The study also remains industry-sponsored, and includes interpretive comparisons to other interventions (for example: surgery, spacers) that are not based on direct randomized comparisons, limiting the validity of such claims. As with the parent trial, there is no comparison to standard surgical decompression, which remains a key gap in evaluating clinical positioning. Overall, the 2-year MiDAS ENCORE extension suggests that improvements observed at 1 year may persist in a subset of treated individuals, but the uncontrolled design, attrition, and selective follow-up substantially limit confidence in long-term durability and preclude definitive conclusions about comparative effectiveness or sustained benefit relative to alternative treatments.

While MiDAS ENCORE provides the strongest available evidence suggesting short- to intermediate-term improvement in function and pain with MILD compared with ESI, the findings are best interpreted as evidence of superiority to a limited comparator rather than definitive proof of effectiveness within the broader treatment pathway for lumbar spinal stenosis.

Mekhail (2021) and Mekhail (2026) represent related contributions from the same investigative group evaluating the MILD procedure in lumbar spinal stenosis. The 2021 study, a retrospective single-center cohort, reported sustained improvements in pain and reduced need for subsequent surgery over 5 years, suggesting potential durability of effect. The 2026 analysis, based on U.S. Food and Drug Administration (FDA) Manufacturer and User Facility Device Experience (MAUDE) database reports, identified a small number of reported complications ($n=10$), the majority classified as procedure-related (e.g., weakness, numbness, epidural hematoma, dural tear), with few device-related events. Taken together, these findings

suggest that MILD may be associated with symptomatic improvement and a low number of reported adverse events. Strengths include longer-term follow-up in the cohort study and use of a national safety database to characterize potential complications. However, both studies have important limitations. The 2021 study is retrospective, single-arm, and conducted within a single center, limiting causal inference and generalizability. The MAUDE analysis is based on passive, voluntary reporting, lacks a denominator, and cannot estimate complication rates or comparative safety; the very small number of reports further limits interpretability. Additionally, both studies originate from the same research group and are not independent confirmations.

Overall, this body of evidence is insufficient to establish the safety and effectiveness of MILD relative to established alternatives. While suggestive of potential benefit, the data are subject to significant bias and do not provide high-quality comparative evidence needed to support broad medical necessity determinations.

In 2014, the Centers for Medicare and Medicaid Services (CMS) determined that PILD for LSS is not reasonable and necessary under section 1862(a)(1)(A) of the Social Security Act, while authorizing coverage under the coverage with evidence development (CED) framework to support additional research (CMS, 2014). CMS subsequently issued a National Coverage Determination for PILD (NCD 150.13), effective December 7, 2016, covering MILD under CED (CMS, 2016). As of the most recent CMS materials (updated January 2026), PILD/MILD remains covered only under CED, with ongoing approved studies and no removal of the CED requirement.

Percutaneous Posterior Cervical Fusion

Researchers have explored percutaneous posterior cervical nerve root decompression using an expandable intervertebral cervical cage and concomitant posterior arthrodesis as a treatment for single-level cervical radiculopathy due to spondylosis. McCormack (2013) reported the 1-year results of a prospective multicenter single-arm study of 60 individuals treated with the DTRAX Facet System (Providence Medical Technology, Inc., Pleasanton, CA). Neck Disability Index, VAS, and quality of life questionnaire scores were significantly improved at 2 weeks and continued to be significantly improved up to 1 year. At the treated level, 93% of participants demonstrated intrafacet bridging trabecular bone on CT scans. Limitations include the lack of a control group and limited follow-up duration.

Siemionow (2016) reported 2-year clinical and radiographic results for 53 of the original 60 participants. The 2-year outcomes were similar to those reported at 1 year. Radiographic fusion rate was 98.1%. No device failure or surgical reinterventions were reported. As with the initial report, this study was limited by the lack of a control group. Several authors in both studies had consulting relationships with the study sponsors.

Clinical Practice Guidelines

The American Society of Pain and Neuroscience (ASPN) published its *Best Practices for Minimally Invasive Lumbar Spinal Stenosis Treatment 2.0* (MIST 2.0) guideline in 2022 (Deer, 2022). The guideline assigned a Grade A recommendation, supported by Level I-A evidence, for PILD in individuals with LSS, neurogenic claudication, and ligamentum flavum hypertrophy of 2.5 mm or greater. The guideline also included recommendations for individual selection criteria and procedural technique.

The ASPN evidence-based clinical guideline on interventional treatments for low back pain (Sayed, 2022) assigned a Grade A recommendation for PILD and a Grade B, Level I-a recommendation for percutaneous endoscopic discectomy for lumbar disc herniation. The Grade B recommendation for percutaneous endoscopic discectomy was based primarily on studies using direct endoscopic visualization; such procedures involve direct visualization of the operative site and are outside the scope of this document.

Prior ASPN guidance (Deer, 2019) also assigned favorable recommendations for MILD in individuals meeting specific clinical criteria.

Although ASPN assigned Grade A recommendations to PILD and MILD supported by randomized controlled trial evidence, the pivotal trials underlying those recommendations compared PILD to epidural steroid injection rather than to standard surgical decompression. None of the cited trials establish that PILD or MILD produces superior or equivalent outcomes to conventional laminectomy or laminotomy in a head-to-head controlled trial.

The American Pain Society (APS) clinical practice guideline (Chou, 2009) noted insufficient evidence to recommend APLD, intradiscal electrothermal therapy, or nucleoplasty for low back pain or radiculopathy.

The North American Spine Society (NASS, 2020) clinical guidelines and coverage policy recommendations for treatment of low back pain do not support routine use of percutaneous disc decompression techniques as replacements for conventional surgical or nonsurgical management (NASS, 2020; NASS, undated).

The American College of Occupational and Environmental Medicine (ACOEM) practice guidelines for low back disorders also do not support percutaneous disc decompression techniques as standard of care (ACOEM, 2019).

CMS determined that PILD for LSS is not reasonable and necessary, and authorized CED-based coverage (CMS, 2014). CMS has separately determined that laser procedures for disc disease are not covered (NCD 140.5) and that thermal

intradiscal procedures are not covered (NCD 150.11).

Although some studies and guideline statements report favorable findings for PILD/MILD, the evidence remains limited by small sample sizes, lack of blinding, attrition, industry sponsorship, and comparators that do not establish superiority to standard decompressive surgery or optimized conservative management; therefore, the evidence is insufficient to demonstrate improved net health outcomes.

Background/Overview

Spinal surgery is most commonly performed in the cervical and lumbar regions of the spine, which have greater mobility and are therefore more susceptible to degenerative change, misalignment, and instability. Intervertebral disc disease is a frequent cause of spinal symptoms and typically involves herniation of disc material. This may occur when the outer layer of the disc (annulus fibrosus) is disrupted, allowing disc material to protrude or extrude, or the disc may remain intact but stretched, resulting in a contained disc prolapse. These structural changes can compress one or more nerve roots and result in pain, numbness, or weakness. Percutaneous discectomy and disc decompression have been investigated as treatments for back pain associated with disc disease and related structural abnormalities.

Percutaneous techniques using imaging for guidance include automated percutaneous lumbar discectomy (APLD), laser discectomy, and nucleoplasty. APLD involves the percutaneous insertion of a probe into the disc space with fluoroscopic guidance and then physical removal of the disc material using suction curettage. For laser discectomy, a variety of different lasers have been investigated, including yttrium aluminum garnet (YAG), potassium titanyl phosphate (KTP), holmium, argon, and carbon dioxide lasers. Regardless of the type of laser, the procedure involves placement of the laser probe within the nucleus under fluoroscopic guidance. Due to differences in absorption, the energy requirements and the rate of application differ among the lasers. Additionally, it is unknown how much disc material must be removed to achieve decompression. Therefore, protocols vary according to the length of treatment, but typically the laser is activated for brief periods.

The nucleoplasty procedure is similar to the laser procedure but uses bipolar radiofrequency energy in a process referred to as Coblation technology. The technique consists of multiple small electrodes that emit a fraction of the energy required by traditional radiofrequency energy systems. The result is that a portion of nucleus pulposus tissue is ablated not with heat, but with a low-temperature plasma field of ionized particles. These particles have sufficient energy to break organic molecular bonds within tissue, creating small channels in the disc. The proposed advantage of this Coblation technology is that the procedure provides for a controlled and highly localized ablation, resulting in minimal thermal damage to surrounding tissue. Complications following percutaneous disc procedures include reherniation, disc instability, and device malfunction.

Image-guided minimally invasive lumbar decompression, also known as MILD, is a percutaneous image-guided procedure that uses contrast media to enhance visualization during treatment of lumbar spinal stenosis. According to the manufacturer, the device is designed to access the interlaminar space via a posterior approach, allowing removal of small portions of the lamina and targeted resection and debulking of hypertrophied ligamentum flavum to achieve decompression. This procedure does not involve a discectomy. The procedure can be performed on an outpatient basis under local anesthesia (Vertos Medical mild Device Kit; Vertos Medical, Inc., San Jose, CA).

The X-Sten MILD Tool Kit received original U.S. Food and Drug Administration (FDA) 510(k) clearance in 2006. The kit contains a set of specialized surgical instruments used to perform lumbar decompressive procedures for the treatment of various spinal conditions. In 2010, the FDA granted 510(k) clearance (K093062) for the modification of the X-Sten MILD Tool Kit (Vertos, San Jose, CA). According to the FDA clearance summary, the Vertos Medical mild Device Kit is substantially equivalent to the X-Sten MILD Tool Kit (K062038) and the Baxano Ultra Low Profile Rongeur and Access Tools (Baxano, Inc., Mountain View, CA; K062711), which had been granted FDA 510(k) clearance at an earlier date. According to the Vertos MILD instructions for use, the device should be used for tissue resection at the perilaminar space, within the interlaminar space and at the ventral aspect of the lamina. The device is not intended to be used to remove the spinal disc.

Additional percutaneous discectomy devices have also been cleared by the FDA through the 510(k) substantial equivalence process, including the DeKompressor Percutaneous Discectomy Probe (Stryker Instruments, Kalamazoo, MI), Herniatome Percutaneous Discectomy Device (Gallini Medical Devices, South Euclid, OH), and the Nucleotome[®] Probe Set (Clarus Medical, LLC, Minneapolis, MN). The FDA indications for these products state: "For aspiration of disc material during percutaneous discectomies in the lumbar, thoracic and cervical regions of the spine."

Percutaneous posterior cervical nerve root decompression using an expandable intervertebral cervical cage has been explored as an alternative surgical treatment for single-level cervical radiculopathy. According to information on the FDA website, the PMT Cervical Cage (Providence Medical Technology, Inc., Pleasanton, CA) is an intervertebral fusion device intended to be used in cervical spinal fusion surgery. The FDA 510(k) clearance states that the device is intended to be used as follows:

"PMT Cervical Cage is indicated for use in skeletally mature patients with degenerative disc disease (DDD) of the cervical spine (C3-C7) with accompanying radicular symptoms at one disc level. DDD is defined as discogenic pain

with degeneration of the disc confirmed by patient history and radiographic studies. Patients should have received at least six weeks of non-operative treatment prior to treatment with the device. Devices are intended to be used with autogenous bone graft and supplemental fixation, such as an anterior plating system" (FDA Providence Cervical Cage, 2013).

Definitions

Annuloplasty: A minimally invasive procedure that applies thermal energy to the annulus fibrosus of an intervertebral disc to treat discogenic low back pain. The procedure works through thermal heating of the annulus to denature collagen fibers and denervate posterior annular nerve fibers that are the source of pain in degenerative discs.

Annulus fibrosus: The outer fibrous ring of the intervertebral disc, consisting of concentric layers (lamellae) of collagen fibers arranged in alternating oblique orientations that surround and contain the central nucleus pulposus.

Automated percutaneous lumbar discectomy (APLD): A minimally invasive procedure that uses a specially designed automated suction-cutting device to mechanically remove nucleus pulposus material from the center of the intervertebral disc space, thereby achieving intradiscal decompression for the treatment of contained lumbar disc herniation. The technique applies the principle of suction cutting to aspirate disc material percutaneously.

Chemonucleolysis: An injection of a drug to degrade the disc and reduce intradiscal pressure.

Coblation (also known as controlled ablation, cold ablation, cool ablation, ionized field ablation, plasma-mediated ablation, or low-temperature plasma excision): A surgical technique that uses bipolar radiofrequency energy passed through a conductive medium (such as isotonic saline) to create a precisely focused plasma field that breaks organic molecular bonds, causing tissue dissolution at relatively low temperatures of 40°C to 70°C. This contrasts with traditional electrosurgery, which operates at temperatures exceeding 100°C.

Contained versus non-contained disc herniation: Contained disc herniation refers to a herniation in which the displaced disc material remains within an intact outer annulus fibrosus and posterior longitudinal ligament, whereas non-contained disc herniation describes herniation where the disc material has penetrated through the annulus fibrosus or posterior longitudinal ligament, or both (extrusion or sequestration).

Curettage: The removal of tissue using a small, spoon-shaped device called a curette.

Disc decompression: Reducing pressure within the disc by reducing the volume of nuclear material inside the disc, typically by removal, ablation, or alteration of nucleus pulposus material.

Disc degeneration: Age-related or pathologic changes in the intervertebral disc characterized by loss of water content of the nucleus (the center of the disc) and decreased height of the disc, annular fissures (cracks in the outer fibers), and circumferential enlargement of the disc .

Discectomy: Surgical removal of a part or all of an intervertebral disc, with or without removal of the nucleus pulposus, typically performed to relieve nerve root or spinal cord compression. Techniques may include open, microscopic, or minimally invasive approaches.

Discogenic pain: Pain originating from the intervertebral disc itself, typically without nerve root compression, and thought to arise from structural disruption or inflammation within the disc.

Epidural steroid injection (ESI): A minimally invasive procedure that delivers corticosteroid medication into the epidural space surrounding the spinal cord and nerve roots to reduce inflammation and relieve radicular pain.

Herniated disc: Displacement of intervertebral disc material, nucleus pulposus material, or both beyond the normal boundaries of the intervertebral disc, which may include protrusion, extrusion, or sequestration and can result in nerve root or spinal cord compression.

Intradiscal electrothermal therapy (IDET): A minimally invasive percutaneous procedure that involves heating an intradiscal probe through electrical current to treat chronic discogenic low back pain. The procedure uses a steerable thermal resistance wire placed along the posterior annulus fibrosus, with thermal energy applied to destroy penetrating nociceptive nerve fibers and modify collagen cross-linking to stiffen the intervertebral disc.

Lamina: A thin broad plate of bone forming the most posterior portion of the neural arch. The two laminae of each vertebra meet in the midline to form the spinous process.

Laminectomy: Surgical removal of a spinal lamina, often done to decompress underlying spinal nerves or to provide access for another surgical procedure.

Laminotomy: Partial removal of the spinal lamina to provide access to the neural canal.

Ligamentum flavum: Elastic ligaments connecting adjacent laminae that contribute to spinal stability; hypertrophy of these ligaments is a common contributor to spinal stenosis.

Lumbar stenosis: A narrowing of the lumbar spinal canal, lateral recess, or neural foramina in the lumbar spine, which may result in compression of neural elements and symptoms such as neurogenic claudication or radiculopathy.

MacNab Clinical Outcome Measures:

- Excellent - No pain; no restriction of activity;
- Good - Occasional back or leg pain of sufficient severity to interfere with the ability to do normal work or capacity to enjoy leisure activities;
- Fair - Improved functional capacity but handicapped by intermittent pain of sufficient severity to curtail or modify work or leisure activities;
- Poor - No improvement or insufficient improvement to enable increase in activities; further operative intervention required.

Microdiscectomy (also called lumbar microdiscectomy or microscopic discectomy): A surgical procedure performed under microscope magnification to remove the portion of a herniated intervertebral disc that is compressing the nerve root or spinal cord, thereby relieving radicular pain and neurological symptoms.

Minimal clinically important difference (MCID): The smallest change in a participant-reported outcome measure (PROM) score that individuals perceive as beneficial and that would mandate a change in clinical management.

Minimally invasive lumbar decompression (MILD): A percutaneous, fluoroscopy-guided procedure that treats lumbar spinal stenosis by removing portions of the lamina and debulking hypertrophied ligamentum flavum to restore space in the central spinal canal. The procedure is performed as an outpatient intervention under moderate sedation without direct visualization of the surgical area, using specially designed instruments introduced through small incisions.

Neurogenic claudication: A clinical syndrome characterized by pain, discomfort, numbness, tingling, weakness, or fatigue in the buttocks, thighs, or lower legs that is precipitated by walking or prolonged standing and typically relieved by sitting or lumbar flexion. It is the most common and important clinical feature of lumbar spinal stenosis.

Nucleoplasty (also known as percutaneous disc nucleoplasty): A minimally invasive procedure for disc decompression that uses radiofrequency energy to ablate and coagulate nucleus pulposus tissue through a percutaneous approach. The technique employs Coblation technology, which delivers radiofrequency energy through a 1 mm bipolar probe to generate a highly focused plasma field around energized electrodes, breaking organic molecular bonds and creating small channels within the nucleus pulposus tissue.

Nucleus pulposus: The central gelatinous core of the intervertebral disc, composed of a highly hydrated proteoglycan-rich matrix containing chondrocyte-like cells (nucleopolpocytes) that provides compressive load support and shock absorption for the spine.

Oswestry Disability Index (ODI): A self-administered, condition-specific participant-reported outcome measure (PROM) used to quantify the extent of functional disability related to low back pain, consisting of 10 items scored from 0-5 with a total score expressed as a percentage (0-100), where higher scores indicate greater disability. The MCID for the ODI varies depending on the calculation method and study population but an international consensus recommends an MCID of 10 points or 30% improvement from baseline.

Percutaneous: Access through the skin (puncture as opposed to "open" surgical incision).

Percutaneous laser disc decompression (PLDD): A minimally invasive procedure that uses laser energy to vaporize a small portion of the nucleus pulposus of an intervertebral disc, thereby reducing disc volume and intradiscal pressure to relieve radicular pain associated with disc herniation. The technique involves percutaneous introduction of an optical fiber into the intervertebral disc under imaging guidance (typically fluoroscopy, computed tomography (CT), or ultrasound), followed by administration of laser energy to ablate disc tissue.

Radicular pain: A type of pain that radiates to the upper or lower extremity directly along the course of a spinal nerve root. Radicular pain is caused by compression, inflammation, or injury to a spinal nerve root.

Radiculopathy: A clinical syndrome characterized by the presence of weakness, loss of sensation, or loss of reflexes associated with dysfunction of a specific spinal nerve root, or a combination of these findings. It is distinct from, but often coexists with, radicular pain.

Spine anatomy: The spine is divided into four major regions: cervical (neck), thoracic (mid-back), lumbar (lower back), and sacral. These regions consist of vertebrae separated by intervertebral discs. Together, these structures support load-bearing and mobility and protect neural elements.

Surgical approaches:

- Percutaneous - Access via needle or small incision using imaging guidance (no direct visualization).
- Endoscopic - The surgeon uses video or camera guidance during the procedure and performs the surgery through the endoscope working channel.
- Minimally Invasive - The surgeon directly visualizes the anatomy being operated on with the naked eye. The operative field may be developed using small-profile (tubular or other) retraction to limit damage to surrounding normal tissues. The surgeon uses standard microdiscectomy techniques and instrumentation. Visualization may be assisted with a microscope or an endoscope.
- Open - The surgeon directly visualizes the anatomy being operated on with the naked eye. The surgery is performed through an open incision utilizing direct visualization and muscle stripping and retraction. Standard surgical instrumentation and techniques are used and may involve the removal of both bony and ligamentous structures.

Visual analog scale (VAS): A self-reported pain assessment tool consisting of a 10 cm (100 mm) line with descriptive anchors at each end, typically "no pain" on the left and "worst possible pain" or "worst pain imaginable" on the right. The person scoring their pain marks the line at the point corresponding to their current pain intensity, and the score is obtained by measuring the distance in millimeters or centimeters from the left end to the mark. The VAS produces a continuous score from 0 to 100 mm (or 0 to 10 cm). While various cut-off schemes have been proposed, commonly cited interpretations include:

- 0-4 mm: No pain
- 5-44 mm: Mild pain
- 45-74 mm: Moderate pain
- 75-100 mm: Severe pain

Established minimal clinically important difference (MCID) values for VAS include:

- 10 mm (1 cm) on a 0-100 mm scale for acute postoperative pain
- 15 mm (1.5 cm) for chronic low back pain (or 30% reduction from baseline)
- 16.6 mm for emergency department individuals
- A 33% decrease from baseline represents a reasonable standard for meaningful change

Zurich Claudication Questionnaire (ZCQ) (also known as the Swiss Spinal Stenosis Questionnaire): A validated, self-administered, disease-specific participant-reported outcome measure designed to evaluate symptom severity, physical function, and treatment satisfaction in individuals with lumbar spinal stenosis (LSS) and neurogenic claudication. The ZCQ consists of three distinct subscales:

1. Symptom Severity Subscale: Assesses pain characteristics, numbness, weakness, and balance problems
2. Physical Function Subscale: Evaluates walking ability and functional limitations
3. Satisfaction Subscale: Measures participant satisfaction with treatment outcomes (typically administered postoperatively)

Coding

The following codes for treatments and procedures applicable to this document are included below for informational purposes. Inclusion or exclusion of a procedure, diagnosis or device code(s) does not constitute or imply member coverage or provider reimbursement policy. Please refer to the member's contract benefits in effect at the time of service to determine coverage or non-coverage of these services as it applies to an individual member.

When services are Investigational and Not Medically Necessary:

When the code describes a procedure indicated in the Position Statement section as investigational and not medically necessary.

CPT

22899	Unlisted procedure, spine [when specified as a percutaneous procedure for decompression, for example, DTRAX cage procedure]
62287	Decompression, percutaneous, of nucleus pulposus of intervertebral disc, any method utilizing needle-based technique to remove disc material under fluoroscopic imaging or other form of indirect visualization, with discography and/or epidural injection(s) at the treated level(s), when performed, single or multiple levels, lumbar
62330	Decompression, percutaneous, with partial removal of the ligamentum flavum, including laminotomy for access, epidurography, and imaging guidance (ie, CT or fluoroscopy), bilateral; one interspace, lumbar
62331	Decompression, percutaneous, with partial removal of the ligamentum flavum, including laminotomy for access, epidurography, and imaging guidance (ie, CT or fluoroscopy), bilateral; additional interspace(s), lumbar
0274T	Percutaneous laminotomy/laminectomy (intralaminar approach) for decompression of neural elements, (with or without ligamentous resection, discectomy, facetectomy and/or foraminotomy)

any method under indirect image guidance (eg, fluoroscopic, CT), single or multiple levels, unilateral or bilateral, cervical or thoracic

64999 Unlisted procedure, nervous system [when specified as percutaneous decompression or laser procedures of cervical or thoracic spine]

HCPCS

C2614 Probe, percutaneous lumbar discectomy

S2348 Decompression procedure, percutaneous, of nucleus pulposus of intervertebral disc, using radiofrequency energy, single or multiple levels, lumbar [DISC nucleoplasty]

ICD-10 Procedure

0R533ZZ-0R5B3ZZ Destruction of vertebral disc, percutaneous approach [cervical, cervicothoracic, thoracic or thoracolumbar; includes codes 0R533ZZ, 0R553ZZ, 0R593ZZ, 0R5B3ZZ]

0RB33ZZ-0RBB3ZZ Excision of vertebral disc, percutaneous approach [cervical, cervicothoracic, thoracic or thoracolumbar; includes codes 0RB33ZZ, 0RB53ZZ, 0RB93ZZ, 0RBB3ZZ]

0RN33ZZ-0RNB3ZZ Release vertebral disc, percutaneous approach [cervical, cervicothoracic, thoracic or thoracolumbar; includes codes 0RN33ZZ, 0RN53ZZ, 0RN93ZZ, 0RNB3ZZ]

0S523ZZ-0S543ZZ Destruction of vertebral disc, percutaneous approach [lumbar or lumbosacral; includes codes 0S523ZZ, 0S543ZZ]

0SB23ZZ-0SB43ZZ Excision of vertebral disc, percutaneous approach [lumbar or lumbosacral; includes codes 0SB23ZZ, 0SB43ZZ]

0SN23ZZ-0SN43ZZ Release vertebral disc, percutaneous approach [lumbar or lumbosacral; includes codes 0SN23ZZ, 0SN43ZZ]

ICD-10 Diagnosis

All diagnoses

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Index

Automated Percutaneous Lumbar Discectomy (APLD)

Baxano iO-Flex® System

Cervical Cage

Cervical Deuk Laser Disc Repair

Coblation

Disc Decompression

Discectomy

Disc-FX® System

DTRAX

Herniatome Percutaneous Discectomy Device

Jho Procedure

Laser Discectomy

Minimally Invasive Lumbar Decompression (MILD)

Nucleoplasty

Nucleotome Probe Set

Percutaneous Endoscopic Discectomy

Providence Cervical Cage

Stryker DeKompressor Percutaneous Discectomy Probe

The use of specific product names is illustrative only. It is not intended to be a recommendation of one product over another, and is not intended to represent a complete listing of all products available.

Document History

Status	Date	Action
Reviewed	05/14/2026	Medical Policy & Technology Assessment Committee (MPTAC) review. Revised Description/Scope section. Added "Summary for Members and Families" section. Revised Rationale, Definitions, References, Websites for Additional Information, and Index sections.
Revised	12/18/2025	Updated Coding section with 01/01/2026 CPT changes, added 62330, 62331 replacing 0275T deleted as of 01/01/2026 and revised descriptors for 0274T, 62287.

Revised	05/08/2025	MPTAC review. Removed “and endoscopic” from Title. Removed “or endoscopic” from INV/NMN statement. Revised Rationale and References sections. Revised Coding section, removed CPT 62380 and ICD-10-PCS endoscopic codes.
Reviewed	05/09/2024	MPTAC review. Updated Rationale and References section. Updated Coding section to add 22899 NOC code.
	09/27/2023	Updated Coding section; added HCPCS code C2614.
Reviewed	05/11/2023	MPTAC review. Updated References section.
Reviewed	05/12/2022	MPTAC review. References were updated.
Reviewed	05/13/2021	MPTAC review. References and Index sections were updated.
Reviewed	05/14/2020	MPTAC review. The Background, Index and References sections were updated.
Reviewed	06/06/2019	MPTAC review. Rationale, References, and Websites sections updated.
Reviewed	07/26/2018	MPTAC review. The document header wording updated from “Current Effective Date” to “Publish Date.” Rationale, References, and Websites sections updated.
Reviewed	08/03/2017	MPTAC review. Updated the Description/Scope, Rationale, Background/Overview, References, Websites for Additional Information, Index and History sections.
Revised	01/01/2017	Updated Coding section with 01/01/2017 CPT changes and clarified Description/Scope section.
Reviewed	08/04/2016	MPTAC review. Updated the Description/Scope, Rationale, Definitions, Reference, Index and History sections. Removed ICD-9 codes from Coding section.
Reviewed	08/06/2015	MPTAC review. Updated Rationale, References and History sections.
Reviewed	08/14/2014	MPTAC review. Updated Rationale, References and History sections.
Reviewed	08/08/2013	MPTAC review. Updated document to address minimally invasive lumbar decompression (MILD). Updated Description/Scope, Rationale, Background/Overview, Definitions, References, Index and History sections.
Reviewed	05/09/2013	MPTAC review. References updated.
Reviewed	05/10/2012	MPTAC review. Rationale, Background, Definitions and References updated.
Revised	01/01/2012	Updated Coding section with 01/01/2012 CPT code descriptor changes.
Reviewed	05/19/2011	MPTAC review. Description, Rationale, Background, Definitions and References updated. Updated Coding section with 07/01/2011 CPT and HCPCS changes; removed C9729 deleted 06/30/2011.
Reviewed	02/17/2011	MPTAC review. Description clarified. Additional information added to Rationale and Definitions. References updated. Updated Coding section with 04/01/2011 HCPCS changes.
Revised	04/29/2010	Information regarding the Vertos Minimally Invasive Lumbar Decompression (MILD®) device added to the Rationale. References updated.
Reviewed	02/25/2010	MPTAC review. Coding and references updated.
Revised	02/26/2009	MPTAC review. Position statement revised, title changed, rationale, background, coding and references updated.
Reviewed	11/20/2008	MPTAC review. Updated review date, references and history sections. Updated Coding section with 01/01/2009 CPT changes.
Reviewed	11/29/2007	MPTAC review. Updated review date, rationale, background/overview, references and history sections. The phrase “investigational/not medically necessary” was clarified to read “investigational and not medically necessary.”
Reviewed	12/07/2006	MPTAC review. Rationale and references sections updated.
Reviewed	03/23/2006	MPTAC review.
Revised	11/18/2005	Added reference for Centers for Medicare and Medicaid Services (CMS) - National Coverage Determination (NCD).
Revised	07/14/2005	MPTAC review. Revision based on Pre-merger Anthem and Pre-merger WellPoint Harmonization.

Pre-Merger Organizations	Last Review Date	Document Number	Title
Anthem, Inc.	07/27/2004	SURG.00052	Chronic Spine Pain Treatments/Procedures (Minimally Invasive)
WellPoint Health Networks, Inc.	09/23/2004	3.07.04	Percutaneous Techniques for Disc Decompression

Applicable to Commercial HMO members in California: When a medical policy states a procedure or treatment is investigational, PMGs should not approve or deny the request. Instead, please fax the request to Anthem Blue Cross Grievance and Appeals at fax # 818-234-2767 or 818-234-3824. For questions, call G&A at 1-800-365-0609 and ask to speak with the Investigational Review Nurse.

Federal and State law, as well as contract language, including definitions and specific contract provisions/exclusions, take precedence over Medical Policy and must be considered first in determining eligibility for coverage. The member's contract benefits in effect on the date that services are rendered must be used. Medical Policy, which addresses medical efficacy, should be considered before utilizing medical opinion in adjudication. Medical technology is constantly evolving, and we reserve the right to review and update Medical Policy periodically.

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